Computer Communications 154 (2020) 370–379

Contents lists available at ScienceDirect

Computer Communications

journal homepage: www.elsevier.com/locate/comcom

Cloud providers ranking and selection using quantitative and qualitative approach

Devi R. [a,*], Shanmugalakshmi R. [b]

[a] Department of IT, Government College of Technology, Coimbatore, Tamil Nadu 641013, India[1]
[b] Government College of Engineering & Technology, Karaikudi 630003, Tamil Nadu, India[1]

ARTICLE INFO

Keywords:
Cloud computing
QoS
Cloud service ranking
Service selection
Linear programming

ABSTRACT

Cloud computing is basically an internet based computing, whereby shared configurable resources are provided to cloud service consumers as services on demand. As an increasing growth of cloud computing, many enterprises provide different cloud services to cloud service consumers. From cloud service consumer's perspective, it is difficult to choose an appropriate cloud service that satisfies their QoS requirements. As requirements of one cloud service consumer will vary from another, dynamic ranking has to be used to satisfy the requirements of different cloud service consumers. A simple model is needed to address the dynamic ranking of cloud services. The dynamic ranking and selection of cloud services is solved using Linear Programming (LP) model. This project considers quantifiable attributes such as processor speed, cost, etc. and some non-quantifiable attributes like feedback to rank various cloud services according to the requirements of cloud service consumer using Linear Programming (LP) technique.

1. Introduction

Cloud is an emerging technology which provides utility services as resources to the customers on demand. More number of businesses is utilizing these services. In effect of this more number of cloud service providers exists today in the cloud market, and they are expanding exponentially. Most of the cloud service providers, offers similar functionality, which leads to a selection problem of cloud customers. Selecting the best service for the customers' requirement is a challenging and time consuming job. Hence a framework has been developed in which users list their needs as requirement parameters for which the service providers express their offerings [1]. The QoS parameters such as response time, availability, security etc. are used to rank the cloud services based upon consumer's requirements. Quantitative parameters are objective quality that can be measured, e.g., response time, availability, throughput, etc. Qualitative parameters are based upon the subjective quality of a thing which is based typically on user experience such as service usability [2]. The requirements will vary from one consumer to another consumer. Experience of the existing cloud users is also taken into account in order to select the best cloud service provider.

Problem domain

Cloud computing presents a novel way to enhance the presented consumption and deliverance model for IT services based on the Internet, by providing for sluggishly scalable and often virtualized possessions as a service over the Internet. Cloud computing providers offer their services according to several fundamental models: infrastructure as a service (IaaS), platform as a service (PaaS), and software as a service (SaaS). Enterprises are increasingly moving their IT environments into the cloud, reducing operating costs by converting from a business model reliant on hardware and software ownership, to one based on utility service consumption [1]. The output of this service resulted in deprioritize rise of cloud providers, serve their offerings as a service model. Whereas this also creates additional challenges like QOS, reliability, security, etc. Some of the additional challenges are comparing and monitoring service level agreements of heterogeneous services from different cloud providers, effectively comparing different cloud services based on customer requirements, addressing them with security aspects like access control list, authentication etc., based on vendor's standards. These challenges are carried out by cloud service ranking and selection. Software as a service (SaaS) Enterprises is increasingly moving towards IT. Choosing a trusted cloud provider based on specific service is an open challenge. In IT industries though cloud

* Corresponding author.
E-mail addresses: r.devi@gct.ac.in (Devi R.), drshanmi@gct.ac.in (Shanmugalakshmi R.).
[1] Affiliated to Anna University.

https://doi.org/10.1016/j.comcom.2020.02.028
Received 25 May 2019; Received in revised form 28 January 2020; Accepted 7 February 2020
Available online 22 February 2020
0140-3664/© 2020 Elsevier B.V. All rights reserved.

boom up to a high level, still its back end processing are threatening the cloud customers. Important factor for the cloud customer to believe the service provider is trust. Customer trust the cloud provider based on the SLA (Service Level Agreement). Service provider and cloud customer should agree the SLA to start the service. SLA specify the form of a legal contract signed by provider and consumer with penalties for non-compliance. Reading through the entire SLA documents of different providers for a specific service it is a time consuming process. Current public cloud SLA's are weak (generally providing 99.95%), no guarantee on performance and penalties are small. For example if a person wants to buy an item online then, he undergo the reviews for this item. This shows the trust level of the item which makes the customer to buy the item. Customers are only aware of the service provided by the cloud providers. Questions like, does the Provider provides the level of service it promised? Is the request has faster response? Whether the data confidence has provided? Whether the provider has trusted credentials? These questions are unanswerable in real time. The paper focuses on SaaS (Software as a Service). Issues in SaaS model are, the customer's data have stored at provider's data center with other unrelated information. The customer's information have replicated in different places for high accessibility, which can lead to data compromising. Trust model cannot fit in the mathematical properties like Reflexive, Symmetric, transitive, etc. Linear equations have proposed to evaluate the trust value of different cloud providers for a particular service. Generated trust model based on this linear equations are effective when compared to the existing trust models.

2. Existing methods

A framework to measure the quality and prioritize cloud services

RajkumarBuyya et al. [3] proposed a framework that measures the quality and prioritize cloud services. It computes few quantifiable QoS attributes proposed by Cloud Service Measurement Index Consortium (CSMIC) to rank the cloud services. In the first phase, the ranking of a complex problem is modeled in a hierarchy structure that specifies the interrelation among three kinds of elements, including the overall goal, QoS attributes and their sub attributes and alternative services. The second phase consists of a pair wise comparison of QoS attributes to specify their relative priorities and a pairwise comparison of cloud services based on their QoS attributes to compute their local ranks. In final phase, for each alternative service, the relative local ranks of all criteria are aggregated to generate the global ranking values for all the services. Technique used - analytic hierarchy process (AHP) technique is used for ranking. Advantage — it is used to measure various SMI attributes. Many of these attributes, like usability, accountability vary over time. To rank the cloud services based on these SMI attributes. Disadvantage — though it provides a uniform way to evaluate the relative ranking of cloud services for each type of QoS attribute, non-quantifiable QoS attributes were not considered for ranking. So, it is not compatible with various QoS attributes.

Component ranking by taking the advantages of service candidates

Z. Zheng et al. [4] proposed component ranking by taking the advantages of service candidates. Cloud Rank framework ranks the component using past component usage experiences of other different component users. Hence the locations of users are different the Cloud service quality ranking of a user could not be redirected to another user. So, personalized QoS Ranking for cloud services is proposed to evaluate all the Cloud services at the user-side and rank the Cloud services based on the observed QoS performance. Technique used — it ranks the component using past component usage experiences of other different component users. Advantage — invocations are free. Executing a large number of service invocations is time consuming and resource consuming. Disadvantage — hence the greedy algorithm treats the explicitly rated items and the unrated items equally, it is not compatible with large invocations of service users.

Generic QoS framework for cloud workflow systems

X. Liu et al. [5] proposed generic QoS framework for cloud workflow systems that covers the major stages of a workflow lifecycle. They are QoS requirement specification, QoS aware service selection, QoS consistency monitoring and QoS violation handling. The software services which have higher quality then the QoS constraints can be selected by service selection component. The third component is QoS consistency monitoring at the execution stage. Because of dynamic nature of cloud, workflow execution states need to be kept under constant monitoring and QoS verification.

The monitoring component detects QoS violation and some recovery actions should be taken to handle consistency. Technique used — the framework has QoS requirement specification, QoS selection, QoS monitoring and QoS violation. Advantage — the cost for the development and management of these software components can be reduced. Disadvantage — though it covers all stages of QoS life cycle, QoS metrics are not identified and no mechanism to differentiate consumers based on requirements.

Personalized QoS ranking prediction framework

Jianmin Wang et al. [6] proposed a personalized QoS ranking prediction framework for cloud services, which requires no additional service invocations when making QoS ranking was proposed. In CloudRank1, comparing the QoS values the preference between these two services can be calculated. The difference between preferences values may be treated equally that hurts QoS ranking prediction accuracy. By considering the confidence values of different preference values CloudRank2 was proposed. Similar users whose have higher similarities will get higher confidence values. Technique used — the past usage experiences of other users, the ranking approach identifies and aggregates the preferences between pairs of services to produce a ranking of services. Advantage — The response-time and throughput values of each invocation are recorded. Disadvantage — however cloud rank algorithms have provided the same quality of cloud services and this approach evaluates all the service candidates equally and optimal VM allocation for each service need to be used.

The issue of matching sla parameters to find suitable cloud provider

TejasChauhan et al. [7] proposed the work addresses the issue of matching SLA parameters to find suitable cloud provider for particular application. SLA template contains many parameters like cloud's resources (physical memory, main memory, processor speed etc.) and properties (availability, response time etc.). The first step is to define cloud model that contains cloud resources, properties, resource quantity etc. and requirement model that contains application's required resources and its quantity. The next step is to convert these models to graph structure. Next step is to find Pair-wise Connectivity Graph. Final step is to find initial mapping between two models. Similarity flooding concept is used for initial mapping. Technique used — the matching of SLA templates need to be dynamic and continuous monitoring of quality of service (QoS) is necessary to enforce SLA's, because of dynamic nature of cloud. Advantage — defining a metric for measuring QoS software as a service in cloud computing is easy. It is a huge aggregation of feasible resources like hardware platforms. The model able to manage system resources and that it made the system self — adoptable under dynamic workload. Disadvantage — it only matches service providers published capability with consumer's requirements. Thus it could not be able to track the change in cloud performance because of dynamic nature of cloud.

The trust measurement using Game theory

This method measure the trust based on game theory for social networks where trust degree is determined by three aspects, which are reliability provided by the service, positive effective feedback from customer's or provider's, and recommendation for a particular service credibility [8]. Techniques used to measure the above three aspect are Direct interaction in which reliability provided by the service, positive effective feedback from customer's or provider's is found, Indirect interaction in which recommendation for a particular service credibility is found and Punishment mechanisms are used for calculating the trust factor.

$$Ti = \alpha.STi + \beta.FTi + \gamma.CTi \tag{1}$$

In Eq. (1), Ti is the global trust degree, STi, FTi, CTi are reliability provided by the service, positive effective feedback from customer's or provider's and recommendation for a particular service credibility where α, β, γ are their weights. The merit is the effectiveness of trust measurement is increased. The results may vary as the data changes dynamically is the demerit of this method.

Trust models survey in cloud computing

The basic procedure to make Cloud customer to believe Cloud a trustworthy as traditional services [9]. Trustability of the Cloud resources and Cloud providers can be estimated by feedback collection and assessment of the actual system. Technique used for trust evaluation are Direct method in which the service trust values, service providers are sorted according to the past history of satisfaction and Indirect method in which the survey system monitors the history of service providers over a period of time.

$$Tk = \sum t * k/n,\ 1 \leq k \leq n \tag{2}$$

In Eq. (2), k denotes the system of interest, t be the trust value of the system and Tk is the total trust value of the system k.

Identifying trust in social network based on group trust metric

This method build a straight forward trust model that can benefit social network services, like monitoring access control list, controlling access permission, suggestion of services and related parameters [10]. Trust metric algorithm is used to identify the set of trusted providers among all providers by making smaller the influence of unreliable providers.

$$C(v) = max(d * Wuv * Cu) \tag{3}$$

In Eq. (3), $C(v)$ is the credibility of provider/node v. v, d is the decay factor lies between 0 and 1, Wuv is the weight on the edge from u and v, v is the node.

Compliance based Peers Feedback in Cloud environment

A trust model which enables customers to determine trust of service providers by using three different trust calculation types, interaction based trust, compliance based trust and recommendation based trust [11]. Interaction based trust by surveying the historical interactions of Cloud providers. Compliance based trust by monitoring performance metrics of Cloud providers. Recommendation based trust based on the positive feedbacks of the customers about the service utilized.

$$IBT = Ts/(Ts + Tf) * e\text{^}(Pt - St)Td \tag{4}$$

$$CBT = \sum Mu * Pu \tag{5}$$

$$RBT = \sum Rp * RPp \tag{6}$$

The final trust is based on Eq. (7),

$$Trust = (IBT * PIBT) + (CBT * PCBT) + (RBT * PRBT) \tag{7}$$

In Eqs. (4), (5), (6), (7); Ts is successful transaction; Tf is unsuccessful transaction; Pt is Present time; St is Starting time, Mu is Monitored Parameters; Pu is Precedence among parameters; Rp is recommendation of providers by the customers; RPp is recommendation of provider by the providers as shown in Fig. 1.

Cloud service selection using adaptive learning mechanism

Among available cloud services selecting the most trusted service is a challenging task [12]. Each cloud service broker manages clustered cloud services and performs dynamic cloud service selection strategy using adaptive learning mechanism. Initializing cluster of CBR (Cluster Based Reasoning), Adaptive learning mechanism of clustered CBR (incentive, forgetting and degenerate strategies). Dynamically optimize the cloud service selection and return the best service result to user at low computing cost. Does not provides interactive mechanism between cloud service brokers.

Trustworthiness evaluation by joining Quality of Service and customer recommendation

This model calculates the accuracy of QoS value prediction on quantitative trustworthy attributes, and to estimate the customer recommendation of targeted Cloud service by perception ratings on qualitative attributes [13]. Trustworthiness is evaluated by both objective measurement and subjective perception. This method is applied to evaluate Cloud service trustworthiness only in the offline situation. Dynamic measurement of Cloud service is not available.

Trust evaluation model based on weights and gray correlation analysis

This model is based on combining weights and gray correlation analysis is proposed [14]. Direct trust, recommendation trust, and reputation together form a comprehensive trust, resulting in a more accurate overall trust. Rough set theory and Analytic Hierarchy process (AHP)-are used for the Direct trust. Recommendation trust is calculated by a gray relational analysis.

$$CST(ur, CSk) = \alpha * DT + \beta * RT + \gamma T\theta \tag{8}$$

In Eq. (8), α is weight of Direct Trust; β is weight of Recommended Trust; γ is weight of cloud service credibility; $CST(ur, CSk)$ is Comprehensive trust degree of the user ur to the cloud service CSk. Dynamic trust update mechanism protects against malicious entities and increases user satisfaction and interaction success rate. The system could be further improved by considering more trust dynamic update factors.

User-based trust model

This model helps users to rely on three sources of databases Provider feedback database, Third-party feedback database and User feedback database [15]. Whenever the customer uses a service, the Availability, Reliability, Integrity, Confidentiality and Authentication(ARICA) are calculated. Model produce a service evaluation value and store in the user's database.

$$Tm = (Tv * w1) + (Tvp * w2) + ((1/n)\sum(Tvtpi)) * w3 \tag{9}$$

In Eq. (9), Tv is Trust value component; Tvp is Trust value of provider; $Tvtp$ is Trust value of third parties and N is Number of third parties. ARICA model trust model helps a user to reduce the reliance on the trust value of provider and third-party feedback and increases the dependence on the user trust value.

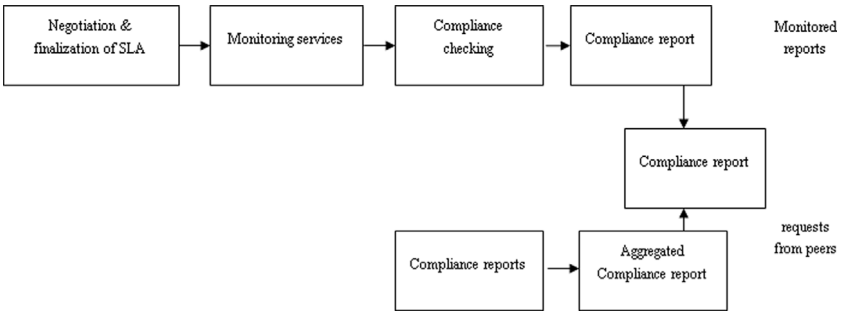


Fig. 1. Compliance based trust calculation.

Trust model based on reputation

The reputation based trust model collects the feedback and opinions from the cloud users and evaluate trust of service provider that is the trust model will select the most reliable and trustworthy CSP by evaluating users feedback [16]. Trust is calculated with the summation of (f, q, r) where, f is Feedback(i)/ Sum_Feedback, q is 1- (Queueload(i)/Sum _Queueload) and r is 1-(Rejection(i)/Sum_Rejection). Feedback rates given to the provider by the user may be falsified

Trust management based on reputation

This Cloud Armor model is a reputation based trust management framework that delivers Trust as a Service (TaaS) and helps distinguish between the credible and the malicious trust feedbacks through a credibility model [17]. Feedback Collusion Detection module protect from malicious users who give fake feedback, Sybil Attacks Detection module protect Cloud services from malicious users who use multiple identities.

$$D(s) = M(s)/(|V(s)|*L(s)) \tag{10}$$

In Eq. (10), $M(s)$ is Total no of users who give feedback; $V(s)$ is Total no of feedbacks and $L(s)$ is feedback volume collusion factor,

$$L(s) = 1 + (\sum |Vc(c,s)|/|V(s)|), |Vc(c,s)| > ev(s) \tag{11}$$

In Eq. (11), $|Vc(c,s)|$ is no of feedback given by users above specified volume collusion threshold.

Trust management framework based on feedback

The trust management framework that can filter out the unreliable feedback based on the behavior of cloud users, ageing factor, majority feedbacks, exogenous method and measure the trustworthiness of service provider [18]. Data Collection module collects user identity, time of transaction, service identity, and service type. Trust Evaluation model verify and avoid ratings related attacks on collected ratings.

$$TV(p) = \sum MF(li,pi)|n|i = 1 \sum TMF(pi)|n|i = 1 \tag{12}$$

In Eq. (12), $MF(li,pi)$ is feedback from ith consumer within new majority threshold set. $TMF(pi)$ is total majority feedbacks between new majority threshold.

A framework of cloud service selection based on trust mechanism

Considering the parameters trust, cost and time, a multi-QoS-aware cloud service selection strategy, and employed the Analytic Hierarchy Process (AHP) method to help the cloud customer to select the cloud service [19]. Analytic Hierarchy Process compute QoS values of cloud services to guide decision makers. Multi-QoS-aware cloud service selection strategy based on trust, cost, and time, are selected to compute the utility value.

$$U = (w1 \times ValTrust) + (w2 \times ValCost) + (w3 \times ValTime) \tag{13}$$

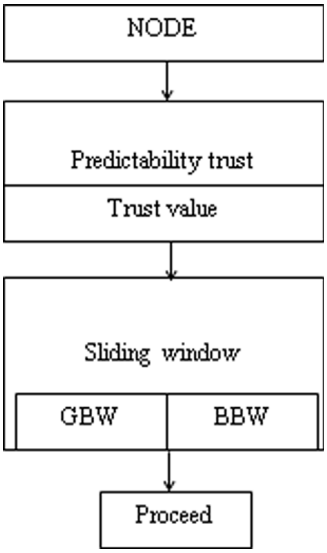


Fig. 2. Trust calculation using sliding window.

In Eq. (13), U is Utility Value; $w1$ is weight of Trust for provider1; $w2$ is weight of Trust for provider2; $w3$ is weight of Trust for provider3. FS = (V1, V2, . . , Vn)^T where, Tr is Trust of cloud service; C is Cost of cloud service; Ti is Time of Cloud service and FS is Final trust score. The change of the cloud service's security is described more precisely and provide a simple and effective decision basis for customers.

Cloud Trust Evaluation model

Fig. 2 shows a trust evaluation model based on evidence theory and sliding windows [20]. Evidence theory mechanism which is highly effective in defending attacks on the system for malicious behavior and sliding windows in which the timeliness of interactions is reflected. |t_curr_t_pos| = Sp; |t_curr_t_unc| = Su; |t_curr_t_neg| = Sn; Where t_curr expresses the current time; t_pos, t_neg and t_unc are the critical time and Sp,Su,Sn represents the window size.

The analytic hierarchy process

This process is introduced by Thomas Saaty (1980). This is an eminent tool for dealing with complex decision making, to assign priorities to the services and finalize the best decision Fig. 3. By comparing pair-wise series which reduces the complexity, and then synthesizing the results, the AHP helps to capture both subjective and objective aspects of a decision. In addition, the AHP incorporates a useful technique for checking the consistency of the decision maker's evaluations, thus reducing the bias in the decision making process [3]. Table 1 show the requirements according to the criterion, where j and

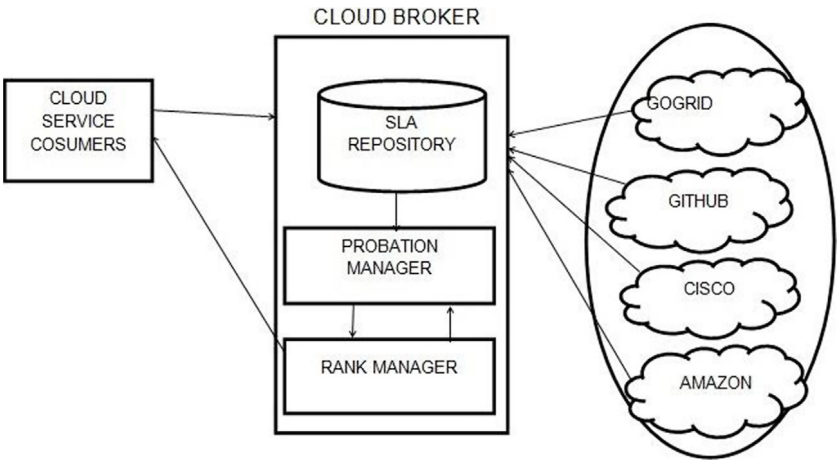


Fig. 3. Analytic hierarchy process.

Table 1
AHP Requirements.

Value of a j k	Interpretation
1	j and k are equally important
3	j is slightly more important than k
5	j is more important than k
7	j is strongly more important than k
9	j is absolutely more important than k

k are priorities from the customer. COMPLEXITY — By increasing the number of services which must be ranked, the execution process tends to need more time. The Complexity of Analytical hierarchy process is O (n3+m3n). Where n is the number of cloud services and m is the number of users.

Ranked voting method

Instead of voting only for a single candidate, voters in Instant Runoff Voting elections can rank the candidates in order of preference. Ballots are initially counted for each elector's top choice. Candidate who secures greater than fifty percent of the votes then that candidate wins. Otherwise the last position candidate is deleted and unconsidered. Remaining places on the ballots are counted again. This process is iterated until there are two choices. Based on "instant runoff" the top two candidates are compared head-to-head. COMPLEXITY *O(C2 log C)*, where *C* is the number of candidates.

3. Proposed method

The proposed architecture has three components Fig. 4. They are Cloud Service Consumers, Cloud Broker and Cloud Service Providers [21]. Various Cloud Service Providers provide similar services to cloud service consumers. So, it is difficult to choose the appropriate cloud service that matches with cloud service consumer's requirements from their perspective. Dynamic cloud service ranking and selection helps cloud service consumers to select the particular services according to their requirements. All cloud service providers register with cloud broker using their SLA's. The Cloud Broker has SLA repository, probation manager and rank manager. The SLA's of cloud service providers are stored in the SLA Repository of cloud broker. When new service provider is added to the provider's pool, then the SLA of new service is stored in the SLA repository.

The Probation Manager takes SLA of new service from SLA repository and checks the parameters of SLA during probation period. After the probation period, the probation manager informs the rank manager with updated parameters. The Rank Manager has rank list that contains ranking of cloud services according to the SLA parameters. The rank manager updates the rank list with SLA parameters given by probation manager. When cloud service consumers request a service from cloud broker with priority of different QoS parameters, the cloud broker modifies the rank list according to each cloud service consumer's requirements and provides an appropriate ranking and selection of cloud services to cloud service consumers.

The reliability and precise of the proposed system is verified through simulation and comparison of the performance with several familiarized schemes.

4. Cloud service provider and consumer functionalities

Build SLA repository

The SLA from various cloud service providers for different cloud services is collected. The SLA form consists of the quantifiable and non-quantifiable QoS parameters which includes service name, cloud provider, availability, processor speed, cost monthly, storage, service credit, response time etc. The SLA parameters from various cloud service providers are collected and stored in the SLA repository. The SLAs are maintained using database. The SLA of cloud service providers are given to cloud broker.

Availability is calculated from the downtime given by the Cloud service provider. Downtime per day is obtained in terms of hours from the provider and the availability percentage is calculated as,

$$Availability\ (\%) = (Uptime/Total\ time) * 100\% \quad (14)$$

$$Total\ time = Uptime + Downtime \quad (15)$$

Processor speed is mentioned in terms of GHz (Giga Hertz). It determines the speed of the processor to perform the task. Cost monthly is given in terms of dollars. It determines how much the provider costs for the usage of his cloud service. Storage space or Memory is given in terms of GB (Giga Bytes). It determines the amount of space given by the cloud service provider. Service Credit is the monthly service fee credit for the affected service. If Provider fails to meet the Service Availability SLA during any one calendar month period, then customer will be eligible for a service credit in the amount of five percent (5%) of the monthly fee for the affected service. For every additional 1% of the calendar month that Service Availability is unavailable, customer will be eligible for credit service which is 5% additional amount for the service fee per month. The affected service percentage can extend up to 50% of service fee per month. The response time criteria specified in SLA should be satisfied. It is defined as the interval of time between the customer initial creation of an incident report (sending SMS or email or online query system) and the provider replies. Different deployment

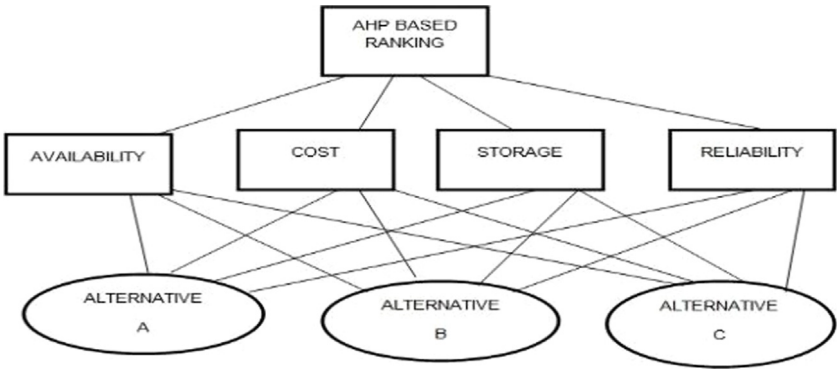


Fig. 4. Ranking and selection architecture.

models supported by cloud are Infrastructure, platform and software services.

Storage is provided as Infrastructure service in which a provider rents its infrastructure as a metered service to other provider or customer to store their data. Database service is provided to query, retrieve and store the contents. It offers scalable, cost effective and time effective to make the business process easy.

Design of cloud broker

The SLA of cloud service providers are collected by cloud broker. It has two entities. They are View Manager and Rank Manager. PROBATION MANAGER — The probation manager checks the SLA parameters and populates the list with SLA. It updates the SLA list whenever a new service enters or when an existing service is updated. After getting SLA parameters, the rank manager performs ranking with the updated SLA list. The provider can also view his SLA details in the view manager. RANK MANAGER — the Rank Manager ranks the cloud services using their SLA parameters. The rank list is updated after new service is added or updated. The rank manager executes the ranking based on the Linear programming Model. The input to the rank manager is given by the customer. The customer specifies the order of requirement on which the SLA list is ranked and then the rank list is given to the customer. It is discussed in the next chapter.

Provider registration

Every provider should register with the application using their SLA details. The provider should create Provider ID and password at the time of registration so that he can login to the system and get access to view and update his SLA details. The SLA details to be filled during registration are Provider name, Downtime per day, Response time, Cost monthly, Storage, Processor speed, Service credit and the Service type. Once the details are submitted, it will be updated in the database for further process.

Provider functions

LOGIN PROVIDER through which the provider can login to the system using his registered Provider ID and password. In case the Provider ID or password is incorrect, the provider cannot perform any action. Once the provider is logged in he can perform the following actions. VIEW PROVIDER — The provider can view his SLA details which he has given at the time of registration. The details such as provider name, availability, response time, Cost monthly, service credit, processor speed and Service type will be displayed. UPDATE PROVIDER — In case the provider wanted to update his SLA details, it can be done through this sub module. The provider can edit any of his details and update it. The updated details will be reflected in the database. The provider can view the updated SLA details.

Customer registration

Every customer should register with the application in order to gain access to find the best provider, to view the provider list and to give feedback. The customer registration is done by filling the details of the customer such as name, Email, Phone, User ID, password. Once registration is complete the customer can login to the application by using his User ID and password provided they are accurately matched with User ID and password given at the time of registration.

Customer functions

LOGIN CUSTOMER — through which the customer can login to the system using his registered User ID and password. In case the User ID or password is incorrect, the customer cannot perform any action. Once the customer is logged in he can perform the following actions. VIEW LIST — The customer should select his required service type as Network as a Service or Storage as a Service or RDS as a Service. Based on the selected service type, the Provider List will be given to the customer. The Provider List consists of the provider name and the SLA details of the provider. PRIORITY SELECTION — The customer selects his requirement from the priority list which contains the SLA parameters. The order of requirement is obtained from the customer in this section in order to perform ranking. Based on this priority best providers will be listed. FEEDBACK — Feedback of the providers will be collected from the customer in this section. The customer gives the feedback based on his satisfaction level with the cloud service provider. The feedback details given by the customer will be updated in the database. Based on feedback of customer, best providers will be listed.

5. Ranking of cloud service providers

Integrate cloud service consumer's requirement with cloud broker

The cloud service consumer gives their requirements to broker after analyzing the available QoS parameters of cloud services maintained by cloud broker. Based on the order of requirements given by consumer, the weight is assigned to the requirements. The highest ordered requirement parameter will get the highest weight which is equal to the number of requirements entered by consumer. The next to highest will get the weight less than the highest. The weight for non-specified parameter by cloud service consumer is 0. The requirements and their corresponding weights is stored in the repository for further rank evaluation.

Cloud service ranking using LP model

The rank manager uses an objective function as a linear equation. Cloud service consumer specified constraint (linear equations) is used to maximize the objective function by Linear Programming [22]. The services with SLA parameters that maximize the objective function are given with higher rank and the highest service or top-N services in the rank list will be given to the cloud service consumer. MAPPING LP WITH PROPOSED SYSTEM — Let $S1, S2, \ldots, Sk$ be the k service providers and $p1, p2, \ldots, pn$ be the n parameters offered by all the service providers. Let $vi_1, vi_2, .., vi_n$, $1 \leq i \leq k$ be the values of parameters. Suppose the consumer chooses m service parameters based on his priority from the given n $(m \leq n)$ parameters. Then these parameters are assigned weights as follows: The weight of the parameters in the order of priority is $m, m-1, m-2, \ldots, 3, 2, 1$ and the remaining n-m parameters will be assigned zero as all the parameters should be considered. Let us calculate the weighted sum $\sum_{j=1}^{n} v_{ij} w_j$ for each service provider Si, $1 \leq i \leq k$. This weighted sum signifies the preference of the consumer. The consumer is allocated the provider with maximum weighted sum. As the consumer should be given a best provider, the decision variables are $S1, S2, \ldots, Sk$. This problem can be formulated as linear programming problem as follows. The objective function is,

$$Maximize\ (S_i), 1 \leq i \leq k \tag{16}$$

where,

$$S_i = \sum_{i=1}^{n} v_{ij} w_i,\ 1 \leq i \leq k \tag{17}$$

Number of parameters chosen by consumer, m ≤ n. Service name given by consumer,

$$S_n \in S_i,\ 1 \leq i \leq k \tag{18}$$

The services with SLA parameters that maximize the objective function are given with higher rank [23,24]. The rank manager uses the weights according to cloud service consumer's interest and performs multiplication of values of parameters and their corresponding weight. Then the weighted sum of for each cloud service is calculated. The service that maximizes the weighted sum which satisfies the above objective function and satisfies all constraints of consumer is selected and given to cloud service consumers using linear programming model. Here is an example based on the constraint given by the consumer. When the consumer gives the requirement as availability and then storage, availability gets highest priority and storage gets the next highest priority. The Fig. 5 graph shows how the objective function is obtained based on the availability and storage. **Case 1:** The current maximum weight value of the service provider S_i is shown and the objective function at this point is obtained Fig. 5. **Case 2:** The loop is executed to find the next maximum weight value Fig. 6. **Case 3:** The feasible solution at the next point is obtained Fig. 7. **Case 4:** The objective function is obtained such that it satisfies all the constraints given by the consumer and the weight value is maximized Fig. 8.

Best providers based on priority basis

In this module, the ranking is performed based on the linear programming model. The order of the requirement is obtained from the customer who needs to use a cloud service. The parameter which has highest order of requirement gets the highest priority which is equal to the number of requirements given by the customer. The second highest order of requirement gets the next priority value decreased by one and so on. The priority value of the non-selected parameter is zero. The weighted sum for the providers is calculated by the product of parameter value and priority value. The best provider is selected based on the rank value (weighted sum) of the providers. The provider that satisfies the constraint of the customer and maximize the objective function will be suggested to the customer as the best providers.

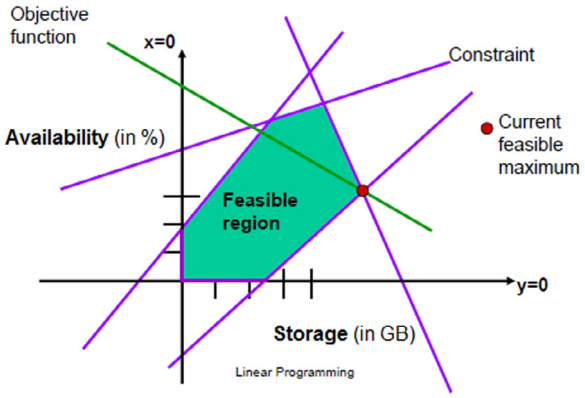


Fig. 5. Objective function for first feasible solution.

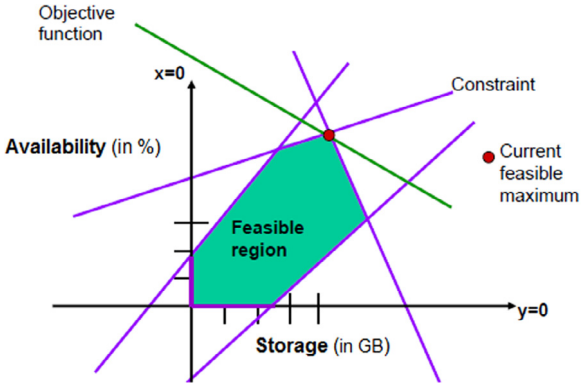


Fig. 6. Objective function for second feasible solution.

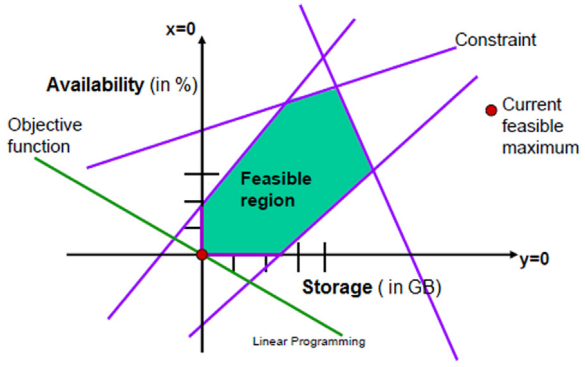


Fig. 7. Objective function for third feasible solution.

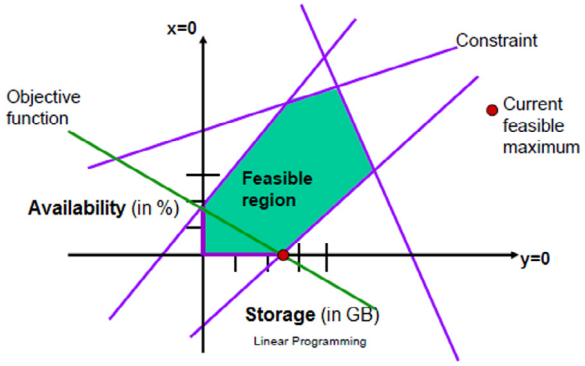


Fig. 8. Objective function for fourth feasible solution.

Best providers based on feedback basis

Every customer gives feedback based on their satisfaction level and based on the given feedback ranking is done. The customer's feedback is collected for each provider and the feedback percentage for each provider is calculated.

$$Feedback\ (\%) = (Positive\ feedback/Total\ feedback) * 100 \quad (19)$$

The provider which has the maximum feedback percentage will become the best provider. The provider with maximum feedback percentage will be suggested when the customer wants to find best provider based on the feedback.

Complexity

The Linear programming algorithm is a P problem. The statements and results are, Problem statement: Maximize the weighted sum based on the customer's requirements. Search version: Input the parameter list of k service providers and the same list is obtained in the order of maximum weighted sum. Verification version: Input the parameter list as X and another list in the order of maximum weighted sum as Y, and output 'YES' if Y satisfies the objective function with respect to weighted sum and 'NO' otherwise. In this case, there are faster solutions to both search version and verification version (a program can be done to add up the values of list X and check for the same value in list Y). Hence Linear Programming is solvable in polynomial time and it is in P class.

6. Experimental results

The SLA document of each cloud service contains quantifiable QoS parameters such as processor speed, cost monthly, storage, service credit, response time, availability and non-quantifiable QoS parameters which include feedback. The cloud service consumers give their requirements to broker, after analyzing the available QoS parameters of cloud services maintained by cloud broker. Based on the order of requirements given by cloud service consumer, the weight is assigned to the requirements. Assuming that the parameters chosen first will have higher precedence and will get the highest weight which is equal to the number of requirements entered by consumer. The following next will get the weight less than the highest. The weight for non-specified parameter by cloud service consumer is 0. The requirements and their corresponding weights is stored in the repository for further rank evaluation. The services with SLA parameters that maximize the objective function are given with higher rank. The rank manager uses the weights according to cloud service consumer's interest and performs product of values of the parameters with their corresponding weight. Then the weighted sum of for each cloud service is calculated. The cloud service that satisfies the objective function is feasible services for consumer. The service that maximizes the weighted sum which satisfies the above objective function with set of all constraints of consumer is selected and given to cloud service consumers using linear programming model. Every customer gives feedback based on their satisfaction level and based on the given feedback ranking is done. The provider which has the maximum feedback percentage will become the best provider. The provider with maximum feedback percentage will be suggested when the customer wants to find best provider based on the feedback. CUSTOMER REGISTRTION TABLE — The customer registration is done by filling the details of the customer such as name, Email, Phone, User ID, password. Once registration is complete the customer can login to the application by using his User ID and password provided they are accurately matched with User ID and password given at the time of registration. PROVIDER REGISTRATION TABLE — The provider should create Provider ID and password at the time of registration so that he can login to the system and get access to view and update his SLA details. The SLA details to be filled during registration are Provider name, Downtime per day, Response time, Cost monthly, Storage, Processor speed, Service credit and the Service type. Once the details are submitted, it will be updated in the database for further process. PRIORITY SELECTION — The customer selects his requirement from the priority list which contains the SLA parameters. The order of requirement is obtained from the customer in this section in order to perform ranking. Based on this priority best providers will be listed. BEST PROVIDERS AFTER RANKING — Those providers which have the maximum weight value will become the best providers [24]. This shows providers satisfies the constraint of the customer and maximize the objective function will be suggested to the customer as the best providers [25]. FEEDBACK BY CUSTOMERS — The customer gives the feedback based on his satisfaction level with the cloud service provider [26]. The feedback details given by the customer will be updated in the database. Based on feedback of customer, best providers will be listed. The Satisfaction level is listed in three categories which would be easily identified by the consumers. The three categories provided are Satisfied, Neutral, and poor. FEEDBACK TABLE — The Feedback details given by the customers for each of the providers is stored in the database. The customer's feedback is collected for each provider and the feedback percentage for each provider is calculated. The provider which has the maximum feedback percentage will become the best provider. Instances of compute service type 1 of different providers are placed in different location. These locations include Ireland, Virginia, Seoul, California and Mumbai. Based on SLA method [7], the availability information of compute service type 1 is collected from the SLA document as shown in Table 2. Using third party tool like cloud harmony availability of Cloud providers are monitored for compute service type 1 and the results are shown in Table 3. Similarly Dynatrace tool is used for proposed method to monitor the compute service type 1 instances of different providers.

Table 2
SLA method.
Source: From SLA document.

Cloud service provider	Compute			
	Service name	Regions	Availability	Last updated
Provider X	Compute service type1	Ireland	99.99%	09-10-2018
		Virginia	99.99%	09-10-2018
		Seoul	99.99%	09-10-2018
		California	99.95%	09-10-2018
		Mumbai	99.95%	09-10-2018
Provider Y	Compute service type1	Ireland	99.99%	09-10-2018
		Virginia	99.98%	09-10-2018
		Seoul	99.97%	09-10-2018
		California	99.95%	09-10-2018
		Mumbai	99.9%	09-10-2018
Provider Z	Compute service type1	Ireland	99.99%	09-10-2018
		Virginia	99.95%	09-10-2018
		Seoul	99.99%	09-10-2018
		California	99.95%	09-10-2018
		Mumbai	99.9%	09-10-2018

By comparing Tables 2–4 proposed method of trust calculation provides optimized approximation compared to other methods. The values shown in these tables are given as input to the linear programming algorithm to identify the feasible solution. Table 4 shows that real time monitoring of instances availability is lesser than specified in the SLA document. Similarly quantitative parameters like feedbacks are collected from the customers, who are using the service. Average of each provider value is given as input to linear programming algorithm to find the feasible solution. Fig. 9 shows how the availability is measured using Dynatrace tool.

7. Conclusion

The goal of the paper is to propose the ranking and selection of cloud services considering both quantifiable and non-quantifiable QoS

Table 3
Agent method.
Source: From third-party website cloudhormony.com.

Cloud service provider	Compute			
	Service name	Regions	Availability	Last updated
Provider X	Compute service type1	Ireland Virginia Seoul California Mumbai	100% 100% 100% 100% 100%	04-02-2019
Provider Y	Compute service type1	Ireland Virginia Seoul California Mumbai	100% 100% 100% 100% 100%	04-02-2019
Provider Z	Compute service type1	Ireland Virginia Seoul California Mumbai	99.99% 99.99% 100% 100% 100%	04-02-2019

Table 4
Proposed method.
Source: From dynatrace tool.

Cloud service provider	Compute			
	Service name	Regions	Availability	Last updated
Provider X	Compute service type1	Ireland Virginia Seoul California Mumbai	100% 98.5% 95.5% 95.5% 91.85%	04-02-2019
Provider Y	Compute service type1	Ireland Virginia Seoul California Mumbai	98.5% 97.3% 100% 94.5% 96.7%	04-02-2019
Provider Z	Compute service type1	Ireland Virginia Seoul California Mumbai	99.4% 93.8% 97.45% 100% 98.4%	04-02-2019



Fig. 9. Provider X average availability measurement using Dynatrace.

parameters to provide an appropriate service that satisfies almost all requirements of cloud service consumers using Linear Programming (LP). To improve the ranking accuracy of these approaches, exploiting additional techniques (e.g., data smoothing, random walk, matrix factorization, utilizing content information, etc.) are used. When a user has multiple invocations of a cloud service at different time, will explore time-aware QoS ranking prediction approaches for cloud services by employing information of service users, cloud services and time. As the current approach only rank different QoS properties independently,

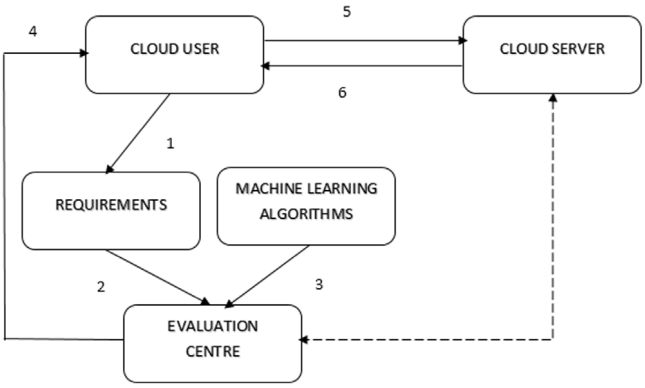


Fig. 10. Trust calculation using machine learning algorithms.

will conduct more investigations on the correlations and combinations of different QoS properties. It will also investigate the combination of rating based approaches and ranking based approaches. So that the users can obtain QoS ranking prediction as well as detailed QoS value prediction. The existing system provides the score according to the static parameters which are considered at the time of evaluation. Here the CC is provided with the score that is available unchanged. Some of the existing system models are Using Centralized model and using De-Centralized model.

New trust evaluation system with machine learning algorithm is used to analyze the trust factors automatically. In this model, algorithm gives the trust value using cloud data trust and reputation of the CSP. We integrate the cloud data trust and reputation values using a combination function of multi-dimensional model to get the overall trust value. There are several ways to combine multi-dimensional trust value. But, it is assumed that both cloud data trust and reputation having equal weight.

The proposed system consists of machine learning algorithms and the information containing the requirements of the CC which is then used to evaluate the trust and reputation value which is send to the CC. This score fully depends on the requirements of the CC. Now according to the specific requirements of the CC, considering the trust value, the selection is made as shown in Fig. 10.

Ethical approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

[1] P. Mell, T. Grance, The NIST definition of cloud computing, Technical Report Version 15, National Institute of Standards and Technology, Information Technology Laboratory, 2009.
[2] M. Subha, M. UthayaBanu, A survey on QoS ranking in cloud computing, Int. J. Emerg. Technol. Adv. Eng. 4 (2) (2014).
[3] J. Saurabh Kumar Garg, Steve Versteeg, Rajkumar Buyya, A framework for ranking of Cloud computing services, Future Gener. Comput. Syst. 29 (2013).
[4] Z. Zheng, Y. Zhang, M.R. Lyu, CloudRank: A QoS-Driven Component Ranking Framework for Cloud Computing, in: Proc. Int'l Symp. Reliable Distributed Systems, SRDS '10, 2010, pp. 184–193.
[5] X. Liu, Y. Yang, D. Yuan, G. Zhang, W. Li, D. Cao, A generic QoS framework for cloud workflow systems, in: Proc. Ninth IEEE Int. Conf. Dependable, Autonomic and Secure Computing, 2011, pp. 713–720.

[6] Z. Zheng, X. Wu, Y. Zhang, M.R. Lyu, J. Wang, QoS ranking prediction for cloud services, IEEE Trans. Parallel Distrib. Syst. 99 (2012) 1, Preprints.
[7] T. Chauhan, S. Chaudhary, V. Kumar, M. Bhise, Service level agreement parameter matching in cloud computing, in: World Cong. ICT, 2011, pp. 564–570.
[8] Yingji Wang, Zhipen Cai, Guishen Yin, Yan Gao, Xiangron Tong, Qilong Han, A game theory-based trust measurement model for social networks, Comput. Soc. Netw. (2016).
[9] K. Gokulnath, Rhymend Uthariaraj, A survey on trust models in cloud computing, Indian J. Sci. Technol. 9 (47) (2016).
[10] Samah Al-Oufi, Heung-Nam Kim, Abdulmotaleb El Saddik, 3.A group trust metric for identifying people of trust in online social network, Expert Syst. Appl. (2012) 13173–13181.
[11] Jagpreet Sidhu, Sarbjeet Singh, Peers feedback and compliance based trust computation for cloud computing, Secur. Comput. commun. (2015).
[12] Xiaogang Wanga, Jian Cao, Yang Xiang, Dynamic cloud service selection using an adaptive learning mechanism in multi Cloud computing, J. Syst. Softw. (2015).
[13] Shuai Ding, Shanlin Yang, Youtao zhang, Changyong Liang, Chenyi Xia, Combining QoS prediction and customer satisfaction to solve cloud service trustworthiness evaluation problems, Knowl.-Based Syst. (2014).
[14] Yubiao Wang, Junhao Wen, Xibin Wang, Bamei Taoand WeiZhou, A cloud service trust evaluation model based on combining weights and gray correlation analysis, Secur. Commun. (2019).
[15] Othman Saeed, Riaz Ahmed Shaikh, A user-based trust model for cloud computing environment, Int. J. Adv. Comput. Sci. Appl. (2018).
[16] Praveen S. Challagidad, Vani S. Reshmi, Mahantesh N. Birje, Reputation based trust model in cloud computing, Internet Things Cloud Comput. (2017).
[17] Talal H. Noor, Quan Z. Sheng, Lina Yao, Schahram Dustdar, et al., Supporting reputation-based trust management for cloud services, IEEE Trans. Parallel Distrib. Syst. (2016).
[18] Sandip Machhi, G.B. Jethava, Feedback based trust management for cloud environment, Inf. Commun. Compet. Strateg. (2016).
[19] Yuli Yang, Xinguang Peng, Donglai Fu, A framework of cloud service selection based on trust mechanism, Int. J. Ad Hoc Ubiquitous Comput. (2017).
[20] Runlian Zhang, Bing Zeng, Shengyuan Zhou, A trust evaluation model for cloud computing, Inf. Technol. Quant. Manage. (2013).
[21] J. Preethi, N. Sujaudeen, T.T. Mirnalinee, P. Venugopal, Cloud service ranking and selection using linear programming, Int. J. Comput. Appl. 124 (3) (2011) 39–43.
[22] Maryam Solhi Lord, Samira MohebbiBazardeh, Linear Programming and Optimizing the Resources, Institute of Interdisciplinary Business Research, 2013.
[23] R. Devi, R. Shanmugalakshmi, Distributed Denial of Service attacks identifying using Ford Fulkerson algorithm in cloud, Int. J. Adv. Technol. Eng. Sci. 5 (1) (2017) 48–57.
[24] R. Dhivya, R. Devi, R. Shanmugalakshmi, Parameters and methods used to evaluate cloud service providers: A survey, in: Computer Communication and Informatics (ICCCI), 2016 International Conference, 2016, pp. 1–5.
[25] R. Devi, R. Dhivya, R. Shanmugalakshmi, Secured Service Provider selection methods in Cloud, in: Advanced Computing and Communication Systems (ICACCS), 2016 3rd International Conference, Vol. 1, 2016, pp. 1–5.
[26] R. Devi, R. Shanmugalakshmi, Design of frame work for cloud provider's selection based on user requirement and trust, Int. J. Sci. Dev. Res. 1 (5) (2016) 313–317.